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Department of Computer Science & Engineering

Report of the Summer Internship

ON

AI & ML Internship

AT

Company Name: Dhanush Softech Pvt. Ltd.

Location: Hyderabad, India

Duration: 30th June 2025 – 29th October 2025

BY

Mr. Malkagalla Karunakar (2451-22-733-085)



Department of Computer Science and Engineering

M.V.S.R. ENGINEERING COLLEGE

(An Autonomous Institution & Recognized by AICTE)

Nadargul, Saroor Nagar Mandal, Hyderabad – 501 510



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Mandal, Hyderabad – 501 510



CERTIFICATE

This is to certify that the summer internship work entitled "AI & ML Internship" is a Bonafide work carried out by Mr.Malkagalla Karunakar (2451-22-733-085) in partial fulfillment of the requirements for the award of degree of Bachelor of Engineering in Computer Science and Engineering from Maturi Venkata Subba Rao (MVSR) Engineering College, Hyderabad during the academic year 2025-2026 under our guidance and supervision.

Internal Guide

N.Sudeepthi Reddy

Assistant Professor

Department of CSE

Head of Department

Dr.Akhil Khare

Professor &Head

Department of CSE



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CERTIFICATE



Date: 6th November 2025

INTERNSHIP CERTIFICATE

This is to certify that Mr. **M. Karunakar** (Int. Code: INT0024) has successfully completed a four-month internship at Dhanush Softech Pvt. Ltd. in the role of an AI & ML Intern from 30th June 2025 to 29th October 2025.

During the internship, he was actively involved in various modules under the following projects:

1. AI Tutor (Q&A)
2. AI Course generator (Delight Feature 1)

Throughout the internship period, he has demonstrated sincerity, commitment, and a keen interest in learning and contributing to the assigned projects.

We wish him all the very best in future endeavours.

For Dhanush Softech Pvt. Ltd.

Chandrima Dendukuri
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DECLARATION

This is to certify that the work reported in the present project entitled "AI & ML Internship" is a record of Bonafide work done by me as part of internship. The report is based on the project work done entirely by me and not copied from any other source.

Malkagalla Karunakar
2451-22-733-085



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ACKNOWLEDGEMENT

I would like to express my deepest sense of gratitude to our esteemed institute M.V.S.R Engineering College (An Autonomous Institution), which has provided us an opportunity to fulfil our cherished desire.

It is with a sense of great respect and gratitude that I express my sincere thanks to my Internal Guide, N.Sudeepthi Reddy Assistant Professor, Department of Computer Science and Engineering, Maturi Venkata Subba Rao Engineering College (A) for the inspiring guidance, supervision, and encouragement towards the successful completion of my internship.

I am also thankful to our Principal Dr. Marthi Kameswara Rao and Dr. Akhil Khare, Professor and Head, Department of Computer Science and Engineering, MVSR Engineering College, Hyderabad for providing excellent infrastructure and a conducive atmosphere for completing this project successfully as part of our B.E. Degree (CSE).

I am sincerely grateful to **Ms. Chandrima Dendukuri, Director, Dhanush Softech Pvt. Ltd.**, for providing me with this internship opportunity and for acknowledging my contributions during the internship period.

Finally, I would like to take this opportunity to thank my family for their constant support throughout this work. I sincerely acknowledge and thank all those who directly or indirectly supported the completion of this work.

Malkagalla Karunakar
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VISION

To impart technical education of the highest standards, producing competent and confident engineers with an ability to use computer science knowledge to solve societal problems.

MISSION

To make the learning process exciting, stimulating, and interesting, To impart adequate fundamental knowledge and soft skills to students, To expose students to advanced computer technologies to excel in engineering practices by bringing out creativity in students, To develop economically feasible and socially acceptable software.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The bachelor's program in Computer Science and Engineering is aimed at preparing Graduates who will:

PEO-1: Achieve recognition through demonstration of technical competence for successful execution of software projects to meet customer business objectives.

PEO-2: Practice life-long learning by pursuing professional certifications, higher education or research in the emerging areas of information processing and intelligent systems at a global level.

PEO-3: Contribute to society by understanding the impact of computing using a multidisciplinary and ethical approach.



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PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Demonstrate competence to build effective solutions for computational real-world problems using software and hardware across multi-disciplinary domains.

PSO2: Adapt to current computing trends for meeting the industrial and societal needs through a holistic professional development leading to pioneering careers or entrepreneurship.

COURSE OBJECTIVES

- To give experience to the students in solving real life practical problems with all their constraints.
- To give an opportunity to integrate different aspects of learning with reference to real life problems.
- To enhance the confidence of the students while communicating with industry engineers and give an opportunity for useful interaction with them and familiarize them with work culture and ethics of the industry.

COURSE OUTCOMES

On completion of this course, the student will be able to:

- Design and develop a small and simple product in hardware or software.
- Complete the task within a pre-specified scope, rather than taking up a complex task and leaving it incomplete.
- Learn to find alternative viable solutions for a given problem and evaluate these alternatives with reference to pre-specified criteria.
- Implement the selected solution effectively.
- Document the work done in a structured and professional manner.



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ABSTRACT

Artificial Intelligence and Machine Learning are transforming modern digital education systems by enabling intelligent learning environments that can automate academic processes, analyze large volumes of educational data, and provide personalized learning support to students. AI-based platforms help learners access knowledge quickly, generate customized study materials, and receive instant assistance through intelligent tutoring systems. These technologies significantly improve learning efficiency, engagement, and accessibility in online education platforms.

The Artificial Intelligence and Machine Learning internship at **Dhanush Softech Pvt. Ltd.**, Hyderabad, conducted from **30th June 2025 to 29th October 2025**, provided practical exposure to the development of intelligent learning modules for educational platforms. The internship involved understanding the architecture of AI-powered systems, integrating machine learning models with backend services, and implementing automated processes for generating educational content and answering user queries. Hands-on experience was gained in designing APIs, managing databases, processing user input, and developing intelligent components capable of interacting with learners.

During the internship, two intelligent modules were developed: **AI Tutor (Question & Answer System)** and **AI Course Generator**. The AI Tutor enables users to ask academic questions and receive instant responses using Natural Language Processing. The AI Course Generator analyzes user interests and learning goals to generate structured course content and personalized learning paths.

This internship enhanced practical knowledge in **Artificial Intelligence, Machine Learning, Natural Language Processing, and intelligent recommendation systems**. The project demonstrates how AI-powered tools can improve digital learning platforms by providing automated tutoring support and personalized course generation, contributing to the advancement of smart educational technologies.

TABLE OF CONTENTS

	PAGE NOs.
Certificate on company letter head	i
Internship Certificate	ii
Declaration	iii
Acknowledgements	iv
Vision & Mission, PEOs, PSOs	v
Course Objectives & Outcomes	vi
Abstract	vii
Table of Contents	viii
List of Figures	ix
Chapters	x

Figure No.	Title	Page No.
Figure 6.1	System Architecture of AI Tutor	08
Figure 6.2	System Architecture of AI Course Generator	08
Figure 6.3	Workflow of AI Tutor System	09
Figure 6.4	Workflow of AI Course Generator System	10
Figure 8.1	AI Tutor and AI Course Generator Dashboard	13
Figure 8.2	n8n Workflow Automation for AI Tutor and AI Course Generator	14

CHAPTERS

1. Introduction	1
2. AI-Based Learning Management Platform	2-3
2.1 Introduction to AI Learning System	
2.2 Objectives of the AI Learning Platform	
2.3 AI & ML Integration in the System	
2.4 System Architecture Overview	
3. Software Technologies Used	4-5
3.1 FastAPI Framework	
3.2 PostgreSQL Database	
3.3 OpenAI API & NLP Models	
3.4 Sentence Transformers	
3.5 n8n Workflow Automation	
4. AI & ML Components Developed	6-7
4.1 Component 01 – AI Tutor (Question & Answer System)	
4.2 Component 02 – AI Course Generator System	
5. Implementation Progress	8-9
6. System Architecture & Workflow	10-12
7. System Implementation	13-14
7.1 AI Tutor System Implementation	
7.2 AI Course Generator Implementation	
8. System Output Screens	15-16
9. Conclusion & Future Scope	17

1. INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) are transforming modern digital platforms by enabling intelligent automation, data-driven decision-making, and personalized user experiences. These technologies allow systems to analyze large volumes of data, identify patterns, and provide meaningful insights that improve the efficiency and usability of applications. AI and ML are increasingly being integrated into domains such as education, healthcare, finance, and e-learning platforms to enhance system functionality and user engagement.

Modern digital learning platforms aim to provide not only educational resources but also intelligent learning assistance. Integrating AI technologies into such platforms enables features such as automated tutoring, smart content generation, and personalized learning recommendations. AI-powered systems can analyze user queries, understand learning needs, and generate structured learning materials, thereby improving accessibility and effectiveness in online education.

This internship was carried out at **Dhanush Softech Pvt. Ltd.**, Hyderabad, where the focus was on developing AI and ML-based components for intelligent learning systems. The internship provided practical exposure to designing intelligent applications, integrating machine learning models with backend services, and implementing automated workflows for data processing and user interaction. The work involved understanding system architecture, developing APIs, managing databases, and implementing Natural Language Processing techniques.

During the internship, two major AI-driven components were developed: **AI Tutor (Question & Answer System)** and **AI Course Generator**. The AI Tutor system allows users to ask academic questions and receive instant responses through Natural Language Processing and machine learning techniques. This system helps students quickly access information and clarify their doubts through an automated tutoring interface.

The second component developed during the internship was the **AI Course Generator**, which automatically generates structured course content based on user interests, skills, and learning objectives. The system analyzes user inputs and creates personalized learning paths, enabling learners to follow customized educational plans suited to their needs.

The primary objective of this internship project was to design and implement intelligent AI-based systems that enhance digital learning platforms. The experience provided hands-on knowledge in Artificial Intelligence, Machine Learning, Natural Language Processing, backend development, and intelligent content generation. Through this work, valuable insights were gained into building scalable AI-powered applications and integrating intelligent features into modern educational platforms.

2. AI-BASED LEARNING MANAGEMENT PLATFORM

2.1 Introduction to AI Learning Management System

A Learning Management System (LMS) is a digital platform used to deliver, manage, and monitor educational content and learning activities. LMS platforms enable institutions and learners to access course materials, track learning progress, and interact with educational resources through an online environment. With the rapid development of technology, modern LMS platforms are increasingly integrating **Artificial Intelligence (AI)** and **Machine Learning (ML)** to provide intelligent learning support and personalized educational services.

AI-powered LMS platforms improve the learning experience by offering automated assistance, personalized recommendations, and dynamic content generation. These systems can analyze user queries, understand learning requirements, and provide relevant educational information. By using Natural Language Processing and machine learning techniques, AI-based systems allow students to interact with learning platforms in a more efficient and intelligent manner.

During the internship at **Dhanush Softech Pvt. Ltd.**, an AI-based learning platform was designed to support intelligent educational services. The system focuses on providing automated academic assistance and personalized course generation using AI technologies. The platform integrates intelligent modules that allow users to ask academic questions and receive instant responses, as well as generate structured learning materials based on their interests and learning objectives.

The platform serves as a centralized digital environment where users can access learning resources, interact with intelligent systems, and receive customized educational guidance. Through AI-powered components, the system enhances accessibility, improves learning efficiency, and supports personalized learning experiences.

2.2 Objectives of the AI Learning Platform

- Providing a centralized platform for accessing educational resources and learning materials.
- Enabling an AI-based tutoring system that allows users to ask questions and receive instant responses.
- Automatically generating structured course content based on user interests and learning goals.
- Using Artificial Intelligence and Natural Language Processing to analyze user queries and deliver relevant information.
- Improving learning efficiency through automated and intelligent educational tools.

These objectives transform traditional learning platforms into intelligent systems that support personalized learning and automated tutoring.

2.3 AI & ML Integration in the System

Artificial Intelligence (AI) and Machine Learning (ML) play a key role in improving the intelligence and functionality of the learning platform. AI-driven systems enable the platform to process user queries, analyze learning needs, and generate personalized outputs that enhance the overall learning experience.

In this system, AI and ML technologies are used to implement intelligent educational tools. One of the main components developed during the internship was the **AI Tutor (Question & Answer System)**, which allows users to ask academic questions and receive instant responses. The system uses Natural Language Processing (NLP) techniques to understand user queries and generate relevant answers.

Another important component is the **AI Course Generator**, which automatically creates structured course content based on user interests, skills, and learning objectives. The system analyzes user inputs and generates personalized learning paths that help users follow a structured educational plan.

By integrating these AI and ML components, the platform becomes capable of providing intelligent tutoring assistance and personalized learning content, improving the effectiveness of digital education systems.

2.4 System Architecture Overview

The AI-based learning platform follows a modular system architecture that separates the frontend interface, backend services, and data management components. This architecture ensures scalability, flexibility, and efficient data processing.

The **frontend layer** provides the user interface through which users interact with the system. It allows users to ask academic questions, access generated course content, and interact with the learning platform.

The **backend layer** handles application logic, API communication, and AI-based processing. Backend services are implemented using frameworks such as **FastAPI**, which supports high-performance API development and asynchronous request handling. The system uses **PostgreSQL** as its primary database to store user information, queries, and generated course data.

In addition, **workflow automation tools such as n8n** are used to manage automated processes and coordinate communication between different services. These workflows help process user requests, trigger AI models, and generate responses or course content automatically.

The integration of AI models, backend APIs, databases, and workflow automation creates a robust system architecture capable of supporting intelligent tutoring and automated course generation within the learning platform.

3. SOFTWARE TECHNOLOGIES USED

During the AI & ML internship, several core software technologies and tools were used to develop intelligent learning components and backend services for the platform.

3.1 FastAPI Framework

FastAPI is a modern Python web framework used for building high-performance APIs. It supports asynchronous programming and enables developers to build scalable backend services efficiently. In this system, FastAPI is used to develop APIs that handle user requests, process AI-generated responses, and communicate with the database.

3.2 PostgreSQL Database

PostgreSQL is an open-source relational database management system used for storing and managing structured data. In this platform, PostgreSQL stores user information, queries, generated course content, and system data. It ensures efficient data storage, retrieval, and management for the AI-based learning system.

3.3 OpenAI API & Natural Language Processing

The OpenAI API is used to enable intelligent interaction between users and the system. It processes user queries and generates meaningful responses using Natural Language Processing (NLP) techniques. In this platform, NLP helps the system understand academic questions and generate accurate answers through the AI Tutor module.

3.4 Sentence Transformers

Sentence Transformers are machine learning models used to generate vector embeddings for textual data. These embeddings allow the system to measure semantic similarity between different pieces of text. In the AI learning platform, sentence transformers can be used to analyze user queries and improve the accuracy of responses and content generation.

3.5 n8n Workflow Automation

n8n is an open-source workflow automation tool used to connect different services and automate processes. In this system, n8n workflows help manage automated tasks such as processing user inputs, triggering AI models, and coordinating communication between APIs, databases, and AI services.

4. AI & ML COMPONENTS DEVELOPED

During the internship, two major Artificial Intelligence and Machine Learning components were developed for the AI-based learning platform. These components were designed to enhance the platform by providing intelligent tutoring support and automated course generation for learners. The two systems implemented were the **AI Tutor (Question & Answer System)** and the **AI Course Generator**.

4.1 AI Tutor (Question & Answer System)

The **AI Tutor** is an intelligent question-and-answer system designed to provide instant academic assistance to users. The system allows learners to ask questions related to various subjects and receive accurate responses generated through Artificial Intelligence and Natural Language Processing (NLP) techniques.

The AI Tutor works by analyzing the user's query and understanding the context of the question using NLP models. The system processes the input text and identifies important keywords and concepts in order to generate relevant responses. By integrating AI models, the system is able to provide meaningful answers and explanations to user queries in real time.

The backend of the system is implemented using **FastAPI**, which handles API requests and processes user queries. When a user submits a question, the request is sent to the backend API where the AI model analyzes the query and generates an appropriate response. The response is then returned to the user through the platform interface.

The system also stores user queries and responses in a **PostgreSQL database**, which helps maintain records of interactions and allows further analysis of user learning patterns. This information can be used to improve system performance and provide better responses over time.

The AI Tutor significantly improves the learning experience by allowing users to quickly resolve doubts and access academic information without manual searching. It acts as an automated digital tutor that supports learners by providing instant knowledge assistance.

4.2 AI Course Generator

The **AI Course Generator** is an intelligent system designed to automatically create structured course content based on user interests, skills, and learning objectives. The platform focuses on simplifying the learning process by generating organized educational material tailored to the needs of individual users.

The system begins by collecting basic information from users such as their preferred subject, skill level, and learning goals. Users interact with the system through a simple interface where they enter the topic or area they wish to learn. The system then processes this input using Artificial Intelligence and Natural Language Processing techniques.

After receiving the user input, the AI model analyzes the topic and generates structured course content that may include key concepts, learning modules, explanations, and suggested study paths. The generated course content is organized into logical sections to help learners understand the subject in a step-by-step manner.

The backend of the system is implemented using **FastAPI**, which handles API requests and processes user inputs. When a user submits a request to generate a course, the backend communicates with the AI model to generate appropriate content. The generated course data can then be stored in a **PostgreSQL database** for future access and management.

The system may also use **workflow automation tools such as n8n** to coordinate interactions between APIs, AI services, and database systems. These workflows help automate the process of content generation and delivery to users.

The AI Course Generator improves the accessibility of education by enabling users to quickly obtain structured learning materials without manual course design. This intelligent system supports personalized learning and helps learners follow a clear educational pathway based on their interests and objectives.

5. IMPLEMENTATION PROGRESS

The development of the AI and ML components for the learning platform was carried out in multiple stages during the internship period. The implementation process involved understanding system requirements, designing system architecture, integrating AI models, and developing automated workflows for data processing. The internship provided practical exposure to building real-world AI-based applications and backend systems.

The first stage of implementation involved studying the platform requirements and identifying areas where Artificial Intelligence and Machine Learning could be integrated. The objective was to enhance the system by introducing intelligent tools capable of providing automated academic assistance and generating structured learning content. Based on these requirements, two main components were selected for development: the **AI Tutor (Question & Answer System)** and the **AI Course Generator**.

The development of the **AI Tutor system** began with designing a question-processing module capable of understanding user queries. Natural Language Processing techniques were used to analyze user input and extract the main concepts from the question. When a user submits a query, the system processes the request through backend APIs and generates a relevant response using AI models. The backend services were developed using **FastAPI**, which allowed efficient API communication and request handling.

User queries and responses were stored in a **PostgreSQL database**, which helped maintain records of interactions and enabled efficient data retrieval. Storing this

information also allows the system to analyze learning patterns and improve response quality over time.

The implementation of the **AI Course Generator** involved developing a system capable of generating structured learning materials based on user interests and learning goals. The system collects input from the user regarding the topic they wish to learn and processes this information using AI models. Based on the input, the system generates course outlines, key topics, and learning modules organized in a structured format.

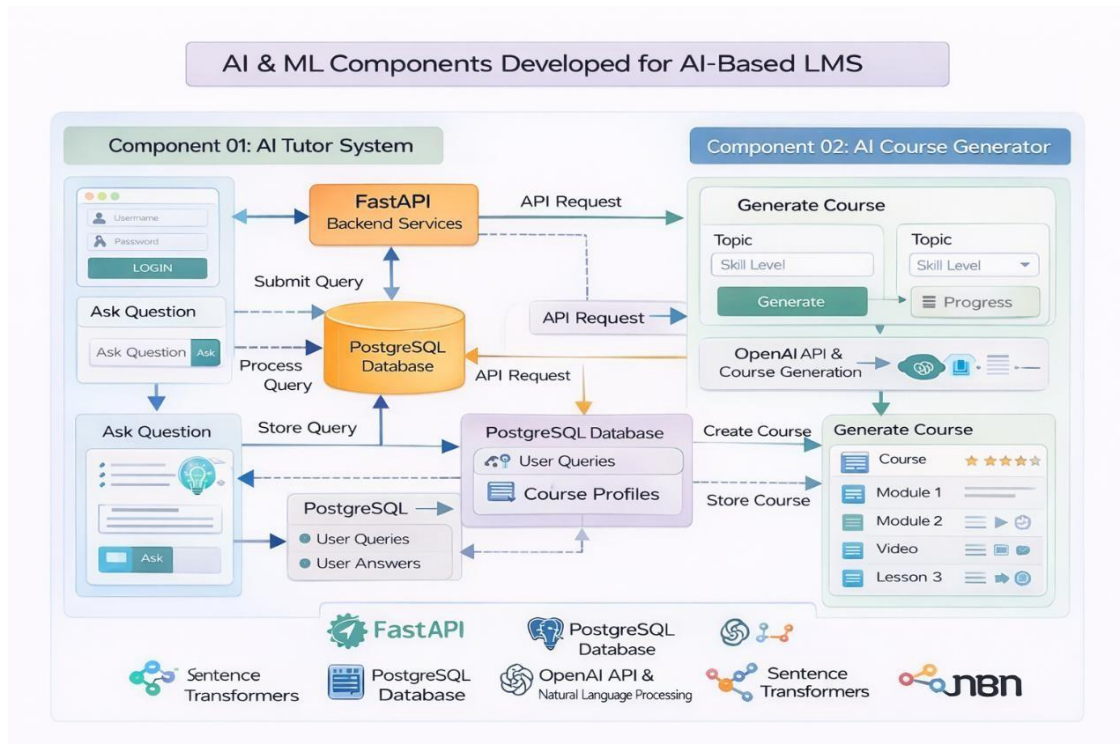
To automate different system operations, **workflow automation tools such as n8n** were used. These workflows help manage communication between APIs, databases, and AI services. The automation process handles tasks such as processing user requests, triggering AI models, and delivering generated responses or course content to users.

Throughout the implementation process, emphasis was placed on writing efficient backend code, designing scalable system architecture, and ensuring proper data management. Continuous testing and debugging were conducted to verify API functionality, database operations, and the accuracy of AI-generated responses.

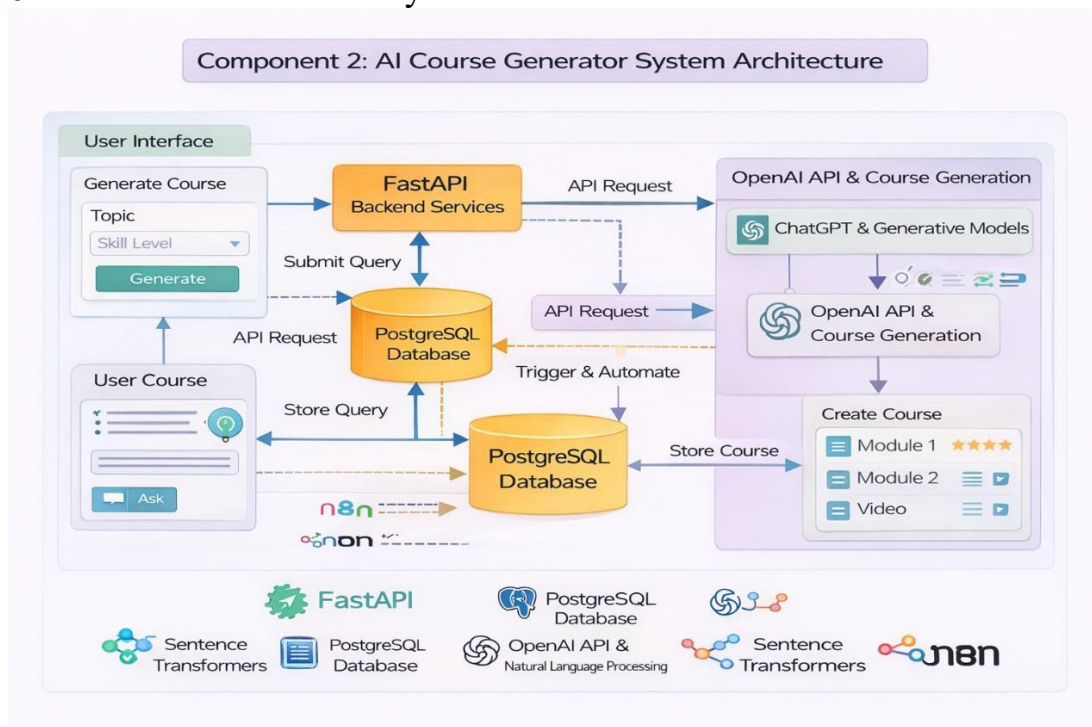
The implementation phase provided valuable experience in developing AI-powered systems and integrating machine learning technologies into real-world applications. By successfully implementing the AI Tutor and AI Course Generator modules, the internship contributed to improving intelligent learning support within the digital education platform.

6. SYSTEM ARCHITECTURE & WORKFLOW

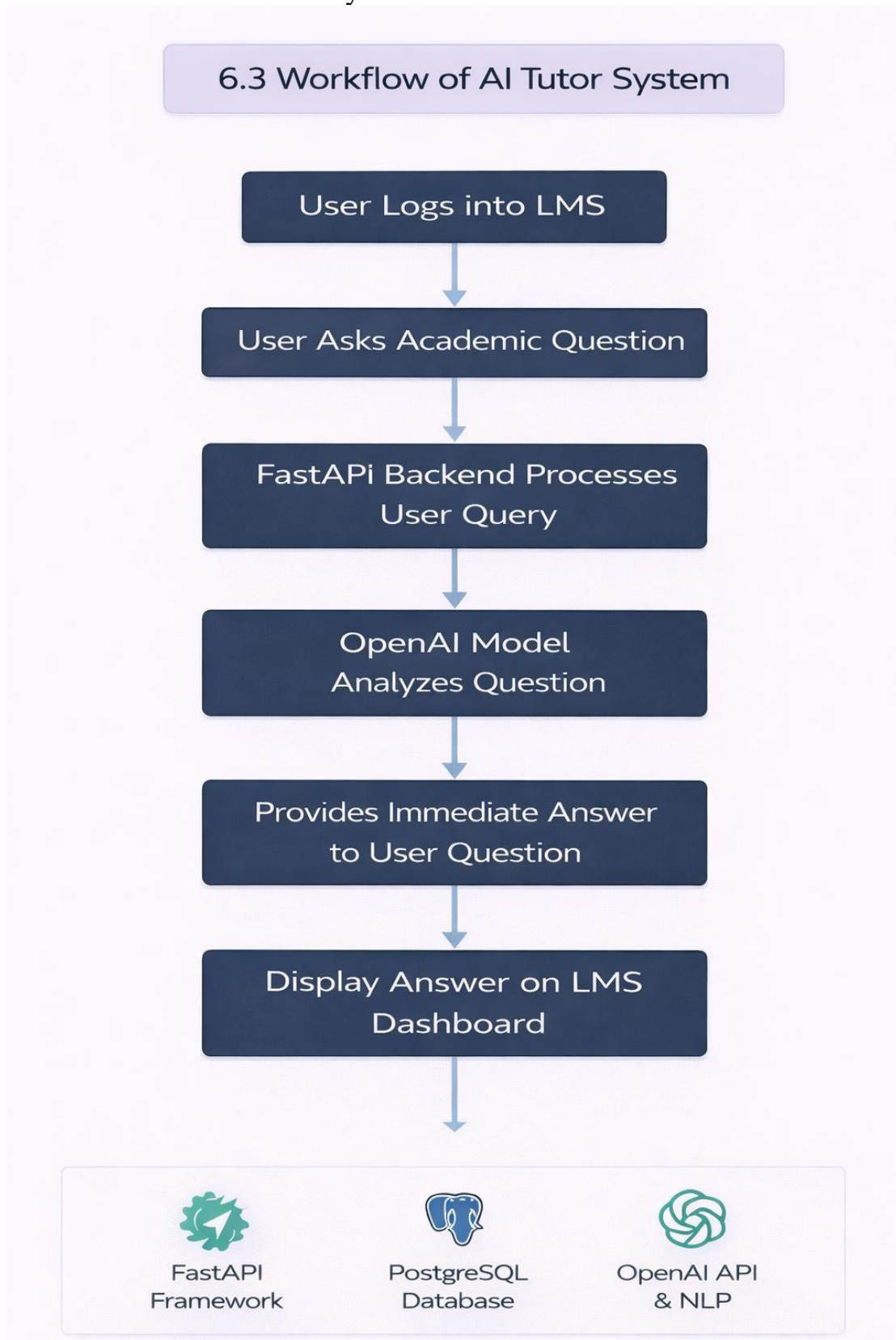
6.1 AI Tutor System Architecture



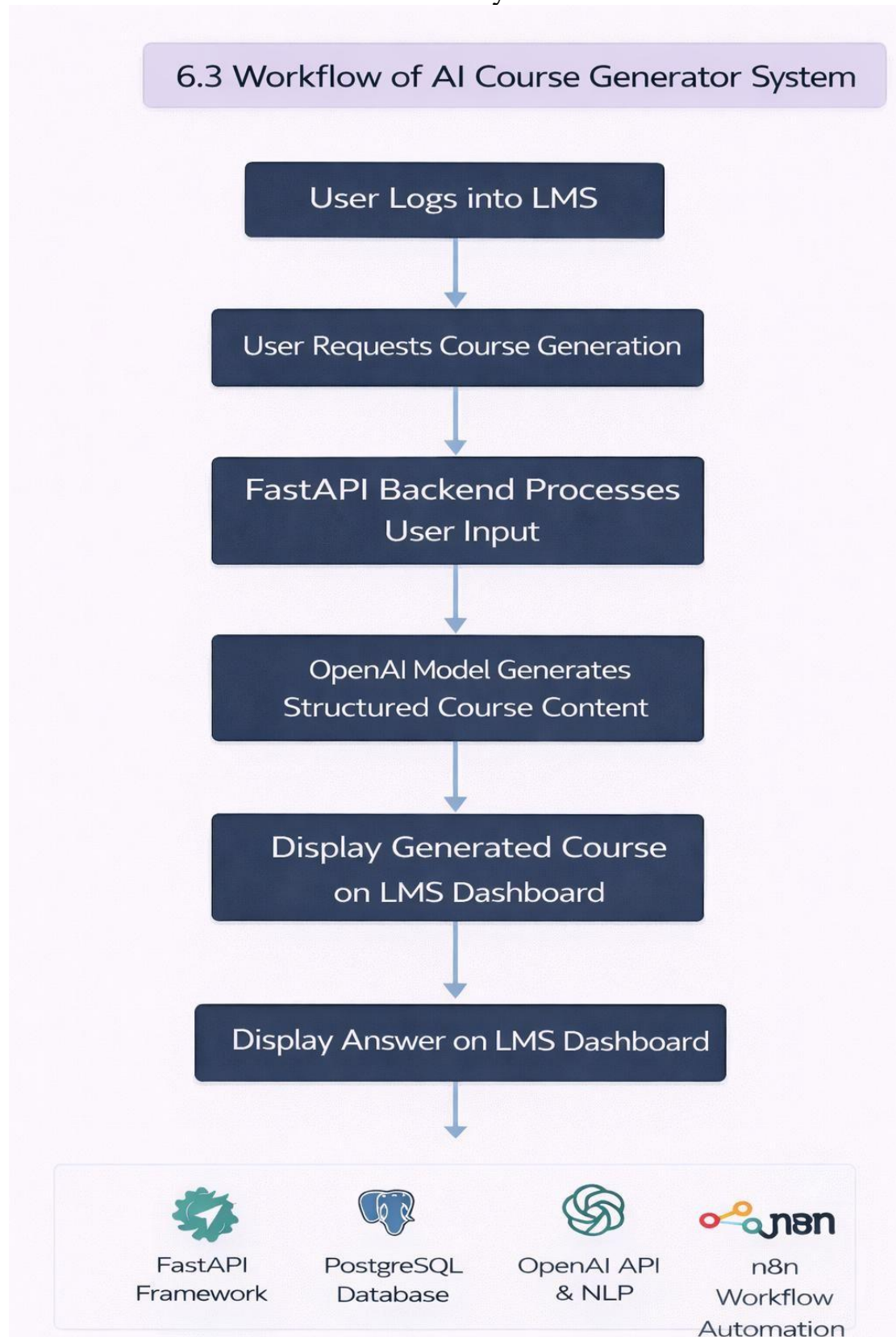
6.2 AI Course Generator System Architecture



6.3 Workflow of AI Tutor System



6.4 Workflow of AI Course Generator System



7. SYSTEM IMPLEMENTATION

This chapter describes the implementation of the Artificial Intelligence and Machine Learning components developed for the AI-based learning platform. The implementation focuses on two major systems: the **AI Tutor (Question & Answer System)** and the **AI Course Generator**. These systems were developed to enhance the platform by providing automated academic assistance and generating structured learning content for users.

The implementation process involved backend development, database integration, AI model interaction, and workflow automation. Technologies such as **FastAPI**, **PostgreSQL**, **OpenAI API**, **Natural Language Processing techniques**, and **n8n workflow automation** were used to develop the system components.

7.1 AI Tutor System Implementation

The **AI Tutor system** was implemented to provide automated academic assistance by allowing users to ask questions and receive instant responses. The system uses Artificial Intelligence and Natural Language Processing (NLP) techniques to understand user queries and generate meaningful answers.

The implementation begins with the user interaction process, where users submit academic questions through the platform interface. The questions are processed through backend APIs and forwarded to the AI processing module for analysis. The system is designed to handle different types of academic queries and provide relevant responses.

Once the query is received, the system analyzes the user input using Natural Language Processing techniques. The AI model processes the text, identifies important keywords, and determines the context of the question. Based on this analysis, the system generates an appropriate response for the user.

The backend services for the system were implemented using **FastAPI**, which enables efficient API communication and request handling. FastAPI processes user requests, communicates with the AI model, and returns the generated response to the user interface.

User queries and generated responses are stored in a **PostgreSQL database**, which helps maintain records of system interactions. The database allows efficient storage and retrieval of data and supports further analysis of user learning patterns.

After the response is generated, the system displays the answer on the platform interface, allowing users to view the solution instantly. This implementation enables the AI Tutor to function as an intelligent digital assistant that helps learners resolve doubts quickly and improve their learning experience.

7.2 AI Course Generator System Implementation

The **AI Course Generator system** was implemented to automatically generate structured learning content based on user interests and learning objectives. The system uses Artificial Intelligence to create organized course outlines that help learners follow a structured educational path.

The implementation begins when users enter a topic or subject they want to learn through the platform interface. This input is sent to the backend server, where it is processed through API requests handled by the FastAPI framework.

After receiving the input, the system performs analysis using AI models and Natural Language Processing techniques. The AI model analyzes the topic and identifies relevant concepts that should be included in the learning material. Based on this analysis, the system generates structured course content.

The generated course content typically includes learning modules, key topics, explanations, and suggested study paths. This structured format helps learners understand the subject in a logical and progressive manner.

The backend of the system was implemented using **FastAPI**, which manages API communication between the platform interface and the AI services. The system also stores generated course information and user inputs in a **PostgreSQL database** for efficient data management.

Workflow automation tools such as **n8n** were used to manage automated processes within the system. These workflows coordinate interactions between APIs, AI services, and database operations, ensuring that course generation and delivery are performed efficiently.

After the course content is generated, the system displays the structured learning materials on the platform interface. This implementation enables users to quickly access personalized learning content and follow organized educational pathways based on their interests.

8. SYSTEM OUTPUT SCREENS

8.1 AI Tutor and AI Course Generator System Dashboards

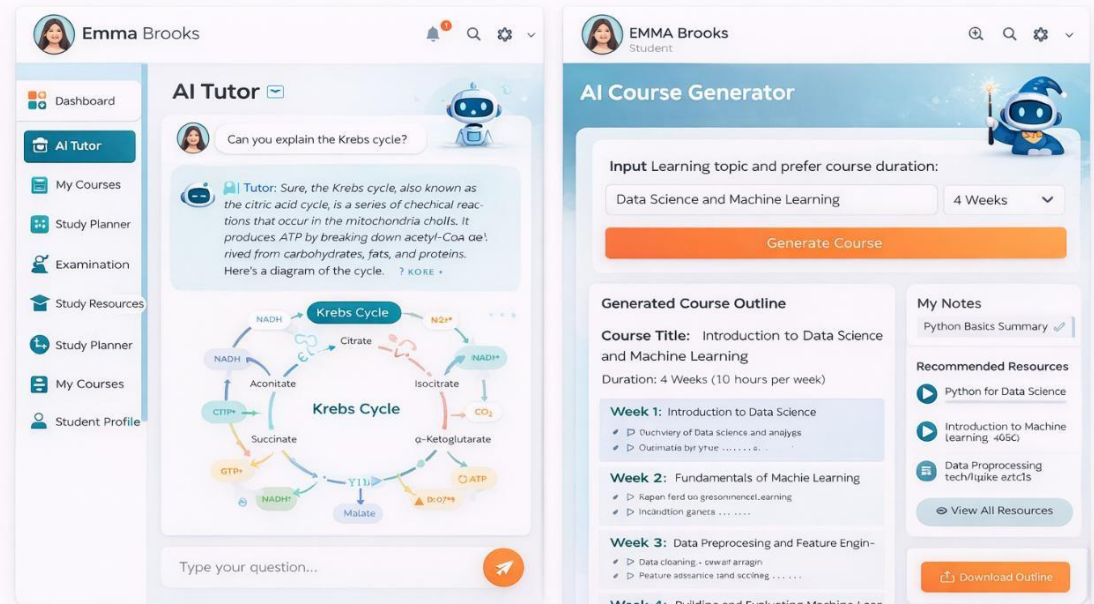


Figure 8.2: Dashboards Showing AI Tutor (left) and AI Course Generator (right)

Figure 8.2: DASHBOARDS SHOWING AI TUTOR (LEFT) AND AI COURSE GENERATOR

Figure 8.1: AI Tutor and AI Course Generator Dashboard

The above interface shows the dashboard of the AI-based learning platform developed during the internship. The dashboard integrates two intelligent modules: the **AI Tutor** and the **AI Course Generator**, which assist users in accessing educational support and generating structured learning materials.

The **AI Tutor module** allows students to ask academic questions through an interactive interface. The system processes the user's query using Natural Language Processing techniques and generates accurate responses in real time. This feature enables learners to quickly resolve doubts and access explanations without manual searching.

The **AI Course Generator module** allows users to enter a learning topic or subject of interest. Based on the provided input, the system analyzes the request using AI models and generates structured course content that includes modules, key concepts, and learning steps. The generated learning materials are displayed on the dashboard, helping users follow a personalized learning pathway.

This dashboard demonstrates how Artificial Intelligence can be integrated into digital learning platforms to provide automated tutoring support and dynamic course generation, thereby improving the overall learning experience for users.

8.2 n8n Workflow Implementation for Shangazi Networking Platform

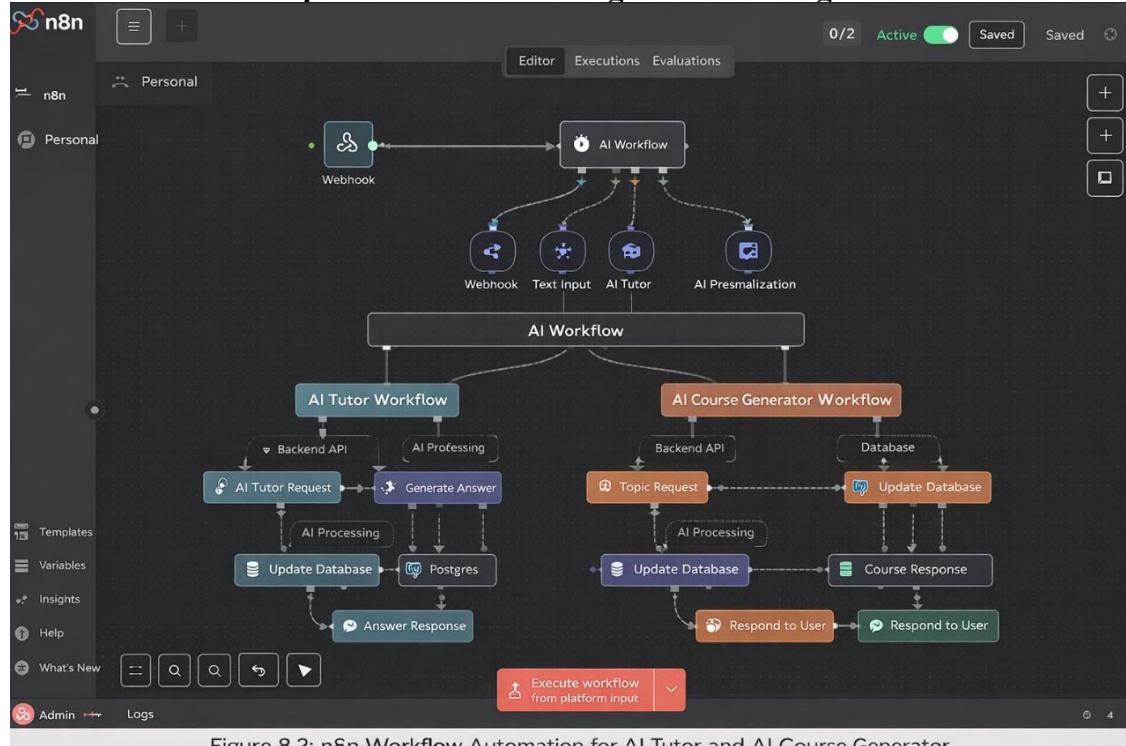


Figure 8.2: n8n Workflow Automation for AI Tutor and AI Course Generator

The above screenshot shows the implementation of workflow automation using the **n8n platform** for the AI-based learning system. The workflow manages communication between the user interface, backend APIs, AI processing modules, and the database.

The process begins with **webhook triggers** that receive user inputs from the platform interface, such as academic questions submitted to the **AI Tutor** or topic requests entered in the **AI Course Generator** module. The workflow forwards these inputs to the backend API for processing.

The backend services interact with the **AI processing module**, where Natural Language Processing techniques analyze the user query or topic. The AI Tutor generates answers to user questions, while the AI Course Generator produces structured course content.

The workflow also performs **database operations**, storing user queries, responses, and generated course materials in the PostgreSQL database. Finally, the results are returned to the user interface and displayed on the dashboard, ensuring efficient processing and intelligent learning support within the platform.

9. CONCLUSION & FUTURE SCOPE

9.1 Conclusion

The internship at **Dhanush Softech Pvt. Ltd.**, Hyderabad provided valuable practical experience in the field of Artificial Intelligence and Machine Learning. The main objective of the internship was to develop intelligent components for a digital learning platform to enhance automated academic assistance and personalized learning support.

During the internship, two AI-based systems were developed: the **AI Tutor (Question & Answer System)** and the **AI Course Generator**. The AI Tutor system allows users to ask academic questions and receive instant responses using Natural Language Processing techniques. This helps learners quickly resolve doubts and access relevant explanations, improving the overall learning experience.

The AI Course Generator focuses on automatically generating structured learning content based on user interests and learning objectives. By analyzing the input provided by the user, the system generates organized course modules, key concepts, and suggested learning pathways that support personalized education.

Through the development of these systems, practical knowledge was gained in **AI integration, backend development, database management, Natural Language Processing, and workflow automation**. The internship provided hands-on experience in building real-world AI-based applications and understanding system architecture in an intelligent educational technology environment.

9.2 Future Scope

Although the implemented systems provide intelligent learning support, there are several opportunities for future improvements. Advanced machine learning models can be integrated to improve the accuracy and relevance of responses generated by the AI Tutor system.

The AI Course Generator can also be enhanced by incorporating adaptive learning techniques that analyze user progress and learning behavior to generate more personalized course content. This would allow the system to recommend learning materials dynamically based on the learner's performance.

Additionally, integrating **real-time recommendation systems, intelligent chatbots, voice-based interaction, and predictive analytics** could further enhance the platform. These improvements would make the system more interactive and transform the learning platform into a fully AI-driven educational environment that supports personalized and efficient digital learning experiences.